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Trust Me, I'm Wrong: LLMs Hallucinate with Certainty Despite Knowing the Answer

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Abstract

Prior work on large language model (LLM) hallucinations has associated them with model uncertainty or inaccurate knowledge. In this work, we define and investigate a distinct type of hallucination, where a model *can* consistently answer a question correctly, but a seemingly trivial perturbation, which can happen in real-world settings, causes it to produce a hallucinated response with high certainty. This phenomenon, which we dub CHOKE (Certain Hallucinations Overriding Known Evidence), is particularly concerning in high-stakes domains such as medicine or law, where model certainty is often used as a proxy for reliability. We show that CHOKE examples are consistent across prompts, occur in different models and datasets, and are fundamentally distinct from other hallucinations. This difference leads existing mitigation methods to perform worse on CHOKE examples than on general hallucinations. Finally, we introduce a probing-based mitigation that outperforms existing methods on CHOKE hallucinations. These findings reveal an overlooked aspect of hallucinations, emphasizing the need to understand their origins and improve mitigation strategies to enhance LLM safety.

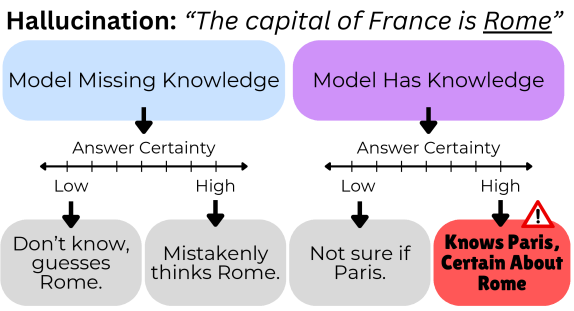

Figure 1: **Do high-certainty hallucinations exist even when the model knows the answer?** An illustrative categorization of hallucinations based on a model's knowledge and certainty. Highlighted is the phenomenon of high-certainty hallucinations (purple) – where models confidently produce incorrect outputs, when they have the correct knowledge. While other types of certain hallucinations can potentially be explained by the model not knowing, or being mistaken, *high-certainty hallucinations, despite knowledge*, are harder to rationalize, making their existence particularly intriguing.

1 Introduction

LLMs often *hallucinate*—generate outputs which are not grounded in real-world facts and may thus hinder their reliability (Ji et al., 2023; Sharma et al., 2023; Kalai and Vempala, 2023). Numerous studies have attempted to identify hallucinations, with a particular line of research highlighting a strong relationship between hallucinations and a model's low certainty (Tjandra et al., 2024; Manakul et al., 2023). These studies demonstrate that certainty estimation metrics (Kuhn et al., 2023; Cole et al., 2023; Feng et al., 2024) can be used to detect and mitigate hallucinations based on an apparent correlation between low certainty and hallucinations.

While low certainty has shown promise for addressing hallucinations, its relationship with hallucinations is not always straightforward. Indeed, recent work suggests that models may hallucinate even when highly certain (Xu et al., 2025; Ji et al., 2025). However, these studies may conflate certainty with lack of knowledge—that is, the model could be certain of a wrong answer simply because it is missing the correct information. In this work, we focus on a distinct failure mode: certain hallucinations that occur even when the model does have the correct knowledge (Figure 1).

Such hallucinations pose serious risks in high-stakes knowledge-intensive domains, where certainty is often taken as a sign of decision reliability, such as medicine (Singhal et al., 2022; Savage et al., 2024), law (Hamdani et al., 2024; Wang et al.,

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2024), and military (Shrivastava et al., 2024).

We term these hallucinations **CHOKE**: **C**ertain **H**allucinations **O**verriding **K**nown **E**vidence. To detect CHOKE examples, we integrate two frameworks into a two-stage procedure: one for identifying hallucinations despite knowledge, and one for estimating model certainty. For the first stage, we build on the approach of Simhi et al. (2024), who identify cases where a model knows the correct answer but hallucinates following a prompt perturbation. Next, we estimate the model's certainty with three widely used but conceptually different methods: tokens probabilities (Feng et al., 2024), probability difference between the top two predicted tokens (Huang et al., 2023), and semantic entropy (Kuhn et al., 2023).

Our findings show that CHOKE examples are widespread, and appear across two datasets with both pre-trained and instruction-tuned models. Furthermore, CHOKE examples exhibit much higher consistency across prompts than other hallucinations, indicating that they form a distinct category.

To evaluate how well existing hallucination mitigation strategies handle these distinct examples, we introduce the CHOKE-Score: a metric tailored to measure mitigation effectiveness specifically on CHOKE examples. Unlike standard metrics, CHOKE-Score isolates performance on examples where the model is both wrong and certain despite knowing the correct answer, revealing failures that may be hidden by overall accuracy. To improve performance on these challenging examples, we introduce a new probe-based mitigation method that focuses the training on CHOKE examples and outperforms existing methods. This evaluation exposes blind spots in current mitigation methods, which is crucial when model certainty guides decisions (Atf et al., 2025; Dahl et al., 2024).

Our contributions are three-fold:

1. We establish that hallucinations can manifest with high certainty despite knowledge of the true answer (CHOKE), challenging the common belief that links hallucinations primarily to low model certainty or inaccurate knowledge.

2. We demonstrate that CHOKE examples show significantly higher consistency across prompts compared to other hallucinations, indicating that they represent a distinct category.

3. We propose *CHOKE-Score*, a novel evaluation metric to assess the effectiveness of hallucination mitigation methods. CHOKE-Score exposes a significant performance gap between overall accuracy and CHOKE examples overlooked by traditional metrics. To address this, we proposed a new probe-based mitigation method that outperforms existing methods.

2 Background

This section overviews related work on uncertainty in LLMs and their tendency to hallucinate, even when the correct answers are known. We rely on these findings throughout our study.

2.1 Uncertainty in LLMs

Predicting the uncertainty of models has been a highly researched topic in NLP and deep learning (Guo et al., 2017; Xiao and Wang, 2019; Gawlikowski et al., 2023). Recent research has explored the origins of low certainty in LLMs, identifying factors such as gaps in knowledge, ambiguity in training data or input queries, and competing internal predictions during decoding (Hu et al., 2023; Beigi et al., 2024; Baan et al., 2023; Yang et al., 2024).

One common application of certainty measures in LLMs is to use them as a proxy to detect hallucinations (Kossen et al., 2024; Wen et al., 2024). This approach is based on the intuition that hallucinations often occur when a model lacks sufficient knowledge to generate a reliable answer, leading to low certainty in its predictions. Studies have shown that abstaining from answering when certainty is low can reduce hallucinations and improve reliability, with minimal impact on cases where a model can generate accurate responses (Cole et al., 2023; Feng et al., 2024).

The simplest approach estimates certainty using the probability assigned to an answer token: the higher the probability, the higher the certainty of the model in its answer. Other methods depend on the model's self-reported certainty in follow-up text generation but are often unreliable (Yona et al., 2024; Beigi et al., 2024). More recent advanced methods consider the full token distribution (Huang et al., 2023) or incorporate semantic similarities across generated tokens (Kuhn et al., 2023).

While prior work has demonstrated that high-certainty hallucinations occur (Xu et al., 2025; Ji et al., 2025), these may result from incorrect or incomplete knowledge. In contrast, our work focuses on a specific subset of hallucinations—cases where

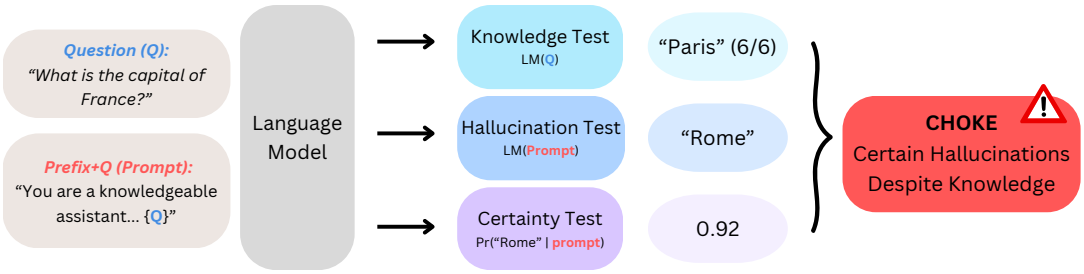

Figure 2: **Detection of CHOKE.** The *Question* is an original dataset question, while the *Prompt* is its subtle variation, simulating real-life natural usage. A sample is classified as CHOKE if all three checks return positive: (a) the model knows the correct answer to the question, (b) it hallucinates an answer when given the natural prompt, and (c) its certainty in its answer exceeds a predefined threshold.

the model has high confidence in an incorrect answer despite being capable of producing a correct answer. This distinction allows us to exclude instances where hallucinations stem from knowledge gaps or incorrect information.

2.2 Hallucination Despite Knowledge

Hallucinations in LLMs have lately become a highly active topic as they impact model reliability (Sharma et al., 2023; Chuang et al., 2023; He et al., 2023; Azaria and Mitchell, 2023; Pacchiardi et al., 2023). Previous work has shown that incorrect or missing knowledge is one of the main reasons for model hallucinations (Béchard and Ayala, 2024; Perković et al., 2024).

That said, recent work found an intriguing phenomenon: hallucinations that occur with prompt variations despite the model possessing the correct knowledge (Simhi et al., 2024; Meng et al., 2024; Bürger et al., 2024; Gekhman et al., 2025; Orgad et al., 2024). These studies differentiate two hallucination types: (1) **lack of knowledge**, where a model does not encode the knowledge, and (2) **hallucination despite having the required knowledge**, where a model generates an incorrect response even when it has the needed knowledge. Our work focuses on the second case of hallucinations, those occurring even when the model knows the correct answer. We leave hallucinations where the model is lacking knowledge for future work.

Identification framework. Specifically, the framework proposed by Simhi et al. (2024) systematically analyzes hallucinations despite knowledge using a three-step methodology. First, they select examples where the model consistently generates the correct answer across multiple generations, including temperature sampling and greedy decoding with a three-shot prompt. Second, they introduce subtle input variations, such as ambiguous phrasing or distractors, to challenge the model's robustness. This input variations approach leverages techniques explored in several studies, which aim to nudge a model toward a mistake (Zeng et al., 2024; Li et al., 2024; Xu et al., 2023; Yao et al., 2023; Nardo, 2023; Joshi et al., 2023; Pacchiardi et al., 2023). Finally, they isolate instances where the model hallucinates under greedy decoding, despite its knowledge. In contrast to Simhi et al. (2024), we show that the CHOKE examples occur even with natural prompts and employing best-practice prompt engineering.

3 Methodology

To show the existence of CHOKE examples, we need to identify them and provide evidence that their portion from the total set of hallucinations is not negligible. To identify CHOKE examples, we use the following procedure: we first identify hallucinations that occur even when the model possesses the required knowledge (Section 3.1). Next, we use common metrics for measuring model certainty (Section 3.2) and set certainty thresholds to separate certain and uncertain generations (Section 3.3). The process of CHOKE examples detection is shown in Figure 2. Additional experimental details are provided in Section 3.4.

3.1 Identifying Hallucinations Despite Knowledge

To isolate hallucinations where the model knows the correct answer, we follow the framework of Simhi et al. (2024). Specifically, we select questions where the model consistently answers correctly using a few-shot prompt, across five temperature-based samples and one greedy decod-